

Seeing the face of the talker improves the intelligibility of speech-in-noise, termed the visual benefit. The discovery that both voice and face contribute to speech perception has spurred thousands of studies in linguistics, psychology, neuroscience, and computer science. The visual benefit is usually measured at the whole-word level, with responses classified as "correct" or "incorrect". An alternative method measures visual benefit using the phoneme overlap between stimulus and response. In 55 participants presented with auditory and audiovisual words, phoneme-overlap analysis estimated 289 different visual benefit values, compared with only three different values for whole-word analysis. Phoneme-overlap detected a positive visual benefit in 67% of word-participant pairs, in contrast to 32% for whole-word analysis. Individual words varied greatly in their phoneme-overlap visual benefit, from 1% to 72%. Variability across words was accurately predicted ($r = 0.78$) by the phoneme composition of the word and the visual benefit for untested words could be predicted with a median error of 9%. Measuring the visual benefit for individual phonemes revealed similarly high variability (1% to 70%) with six phonemes (W, F, V, TH, P, AW) exhibiting visual benefit more than one standard deviation above the mean. The increased sensitivity of phoneme-overlap analysis relative to conventional whole-word analysis may provide a useful tool for investigating audiovisual speech perception in clinical, neuroscience and computer science contexts. For instance, intracranial recordings from human patients have demonstrated the existence of small populations of neurons in temporal lobe that are selective for different groups of phonemes during auditory presentation of speech. In a word-level analysis, observing the face of the talker increased response and decreases response latency in the superior temporal sulcus. If visual benefit varies across phonemes, a phoneme-level analysis might reveal differential modulation of neuronal responses depending on the phoneme content of individual stimulus words. Incorporating expected differences in visual information may also improve the fit of EEG-based methods that rely on a reconstruction of the measured brain signal, capturing variance that otherwise would be assigned to unmodelled error.

C102 Sandbox Series - Tracking overnight lexicalization of non-native phonological contrasts

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Overnight sleep facilitates the lexicalization of newly learned words. For instance, Liu et al. (2024) had participants learn one list of words (non-minimal pairs, differing by multiple phonological units) on Day 1 and another list on Day 2. An identification test was administered following the Day 2 learning. Results revealed faster reaction times (RTs) for Day 1 than Day 2 words, indicating that Day 1 words (learned earlier, remote) had the sleep benefit, whereas Day 2 words (learned before testing, recent) did not. We extend this to minimal pairs in a non-native language (differing by one phonological unit). Cantonese has a more complex tonal and vowel system than Mandarin. Mandarin speakers may struggle with Cantonese phonological contrasts (e.g., Tone 5 (T5)–T6, /ɛ:/-/ɔ:/). Thus, this study investigates whether Mandarin-speaking participants can lexicalize non-native phonological contrasts after an overnight interval. Mandarin-speaking young adults (aged 18–35 years) were recruited (data collection ongoing). Auditory stimuli were

32 Cantonese disyllabic pseudo-words forming eight minimal quadruplets contrasting in tone and vowel (e.g., /pɛ:5fu:1/, /pɛ:6fu:1/, /pɔ:5fu:1/, /pɔ:6fu:1/), paired arbitrarily with 32 everyday objects. Participants learned 16 words on Day 1 and the other 16 on Day 2. Daily training (45 min) comprised one familiarization phase (free listening and viewing, 5 min), after which participants completed separate training for tone and vowel contrasts. For each contrast, they performed fixed learning (word-object matching with feedback; 192 trials; 10 min) and then free learning (clicking an object and hearing "this is [target word], not [the minimal tone/vowel pair of the target word]", 10min). The order of tone and vowel training was counterbalanced across participants. Learning outcomes were assessed by an identification test (256 trials). Independent variables were Type (minimal vs. non-minimal, with minimal further divided into tone and vowel) and Condition (remote vs. recent). A mixed-effects logistic model was analyzed on accuracy, showing a main effect of Type. Accuracy was higher for non-minimal than for minimal pairs ($\beta=1.80$, $SE=0.21$, $z=8.63$, $p<.001$). Within minimal pairs, vowel contrasts yielded higher accuracy than tone contrasts ($\beta=0.79$, $SE=0.18$, $z=4.31$, $p<.001$). For RTs, Type produced a significant main effect ($\beta=-74.27$, $SE=25.31$, $z=-2.94$, $p=.01$), with non-minimal pairs eliciting shorter RTs than minimal pairs (vowel and tone contrasts did not differ from each other). Importantly, the main effect of Condition was significant ($\beta=-58.86$, $SE=16.93$, $z=-3.48$, $p<.001$), with shorter RTs in the remote than recent words, suggesting sleep benefits for both minimal and non-minimal pairs. In the current study, accuracy revealed that minimal pairs were more difficult to learn than non-minimal pairs, with tone contrasts being the hardest. Critically, despite this difficulty, sleep still provided overnight benefits in RTs: remote words (with overnight sleep) showed faster RTs than recent words (without sleep) for both tone and vowel contrasts. Thus, sleep supports language learning even for the most challenging phonological contrasts. These findings advance our understanding of how difficult L2 sounds can be learned at an early stage. We are currently collecting ERP data on lexicalization. A future study will examine another experimental group trained using virtual reality.

C103 Sandbox Series - Does linguistic diversity shape cross-modal event segmentation? ERP correlates of event boundaries in Shipibo-Konibo

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Humans segment unfolding events at different levels of granularity, identifying both coarse (large) and fine (small) event boundaries. Event segmentation has been documented in both visual perception and narrative comprehension and relies on tracking changes across dimensions such as time, space, causality, actors, goals, and objects. Previous research suggests that event segmentation is shaped by external factors such as event familiarity and culture. However, less is known about whether linguistic diversity shapes which event dimensions are tracked across populations during speech comprehension. Here, we investigate whether grammatical marking of event dimensions influences which event boundaries listeners perceive as coarse vs. fine, and whether linguistic event segmentation modulates attention to specific event dimensions during visual event perception. Specifically, we